Polynomial Long Division

Dividing polynomials by long division: linear divisors, dividends with a missing power that needs a zero placeholder, and divisors of degree two or more

How to work these problems.

- Write both polynomials in descending powers, and put a zero coefficient in for every power that is missing. A gap in the columns is where the algorithm goes wrong.
- Each round: divide the leading term of what is left by the leading term of the divisor, multiply the whole divisor by that quotient term, subtract, and bring down the next term. Subtracting means changing every sign.
- Stop when the leftover has lower degree than the divisor. That leftover is the remainder.
- Write the answer as *quotient* + *remainder* over *divisor*, and write *quotient* alone when the remainder is zero.

1. Divide $x^2 + 7x + 10$ by $x + 2$.

Answer: ____________

2. Divide $x^2 + 5x + 9$ by $x + 2$.

Answer: ____________

3. Divide $x^3 - 2x^2 - 5x + 6$ by $x - 1$.

Answer: ________________

4. Divide $2x^3 + 3x^2 - 5x + 4$ by $x + 3$. Then substitute $x = -3$ into the dividend $2x^3 + 3x^2 - 5x + 4$. You should get the same number as your remainder — write it down as a check.

Answer: ________________

5. Divide $x^3 - 8$ by $x - 2$. Write the dividend with every power showing before you start.

Answer: ________________

6. Divide $6x^3 - x^2 - 5x + 2$ by $2x - 1$.

Answer: ________________

7. Divide $2x^3 + 5x - 3$ by $x + 1$. Then substitute $x = -1$ into the dividend $2x^3 + 5x - 3$ as a check on your remainder.

Answer: ________________

8. Divide $x^4 + 2x^3 - 3x^2 + 4x - 5$ by $x^2 + 1$. Remember that the division stops as soon as what is left has degree below 2.

Answer: ________________

9. Divide $x^4 - 5x^2 + 4$ by $x - 2$.

Answer: ____________________

10. Divide $3x^3 - 2x^2 + x - 4$ by $x - 1$. Then check your work: multiply the divisor by your quotient, add your remainder, and write out the polynomial you get.

Answer: ____________________

11. Divide $2x^4 - 3x^3 + x - 5$ by $x^2 - x + 1$.

Answer: ________________

12. Divide $x^3 - 4x^2 + x + 6$ by $x - 3$. The remainder is zero, so $x - 3$ is a factor. Use your quotient to write $x^3 - 4x^2 + x + 6$ as a product of three linear factors.

Answer: ________________